LunarGrid

Distributed Spatial Analytics of the Lunar Surface

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ABSTRACT

The Moon's surface is covered in impact craters, and its poles may hide water ice inside craters that never see sunlight. Where craters are crowded, where the ground is steep, and where light and shadow fall are questions that matter both to planetary science and to choosing safe, useful landing sites. This project, called LunarGrid, brings these ingredients onto a single map and studies them together. It combines three independent crater catalogues (1.3 to 10.7 million craters each), a global laser-altimetry elevation model (118 m per pixel) and polar maps of sunlight and permanent shadow (240 m per pixel).

All the data were aligned in a single spherical reference frame, linked through tile, pixel, and grid-cell indices, and analysed using density maps, hotspot statistics (Getis-Ord Gi*), a strict permutation test, DBSCAN clustering, association rules, and K-means. Apache Spark and Apache Sedona were used to process large volumes of data, distributing spatial work across many processor cores.

The three catalogues showed where the craters are but not how many. Crater density gradually decreased towards the north pole but not towards the south pole. A hotspot, particularly near 70°S, 155°W, appears in all three catalogues even after the strictest test. The permanently shadowed regions hold fewer small craters but more large ones than sunlit ground at the same latitude. Spark and Sedona become more useful when the dataset exceeds around 10 million records, so their value here is in scalability rather than speed on today's data.

Keywords: lunar craters, spatial big data, polar stereographic projection, Apache Spark, Apache Sedona

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1. INTRODUCTION

1.1 Craters, slopes and shadows

The craters help us understand the Moon's history. The areas that have more craters are considered older because they are built up over billions of years. The slope also plays a very important role, as it helps the spacecraft land safely. Shadows matter because some polar craters have not seen sunlight for billions of years, and spacecraft data suggest they may contain water ice, which is a resource for future missions.

Usually, craters, slopes and shadows are studied separately using different data in different map systems. LunarGrid brings them all together as one spatial solution: each is a map layer; the layers must be aligned exactly; and the relationships between them must be measured using statistics that respect the curved geometry of a sphere.

1.2 History of Missing Data

- Many studies use a single crater catalogue, which becomes unsuitable for making decisions about whether a pattern belongs to the Moon or to the catalogue.
- Crater locations, elevation and shadow come in different coordinate systems and with different resolutions, and near the poles a flat map is also badly distorted.
- A grid with equal degrees covers very different ground areas at different latitudes. This can lead to misleading hotspot locations, especially near the poles.
- It is also unclear how much data is needed before using tools like Apache Spark, which becomes more useful than processing everything on a single computer.

1.3 Research Areas to Focus on

The main aim of this project is to understand how crater density, surface slope, and illumination (sunlight) should vary across the Moon, particularly near the polar regions, i.e., the North and South Poles. It also looks at where these features are concentrated and how they relate to one another.

To study this, the project aimed to answer five key research questions:

RQ	Question
RQ1	Where are craters dense, and where do they form clusters or hotspots?
RQ2	How do crater density and size relate to elevation and slope?
RQ3	How do shadow and sunlight vary near the poles, and do craters in shadow differ from craters in light?
RQ4	Do independent crater catalogues lead to the same conclusions?
RQ5	At what data size does distributed processing outperform a single machine?

2. STUDY AREA AND DATA

The study covers the entire Moon, with a closer look at the north and south polar regions at latitudes poleward of 65°. The Moon is considered a sphere with a radius of 1,737.4 km. It has no sea level, so heights are measured relative to this reference sphere and negative values lie below it. The study uses six data layers:

Layer	What it is	Resolution	Used for
Robbins catalogue	Long-standing reference, mapped from LRO imagery	1,296,796 craters ≥ 1 km	Reference positions and sizes
LU5M812TGT	Automatic-detection catalogue (Zenodo)	5,691,533 craters, mostly ≥ 0.4 km	Small craters
LUC-GRAS200	Deep-learning catalogue (CLPDS)	10,699,651 craters, 0.2 to 2,050 km	Secondary-crater labels
LOLA elevation	NASA laser-altimeter global height map	118 m/pixel; about 4.2 billion pixels	Elevation and slope
LOLA polar illumination	Average sunlight and permanent shadow, each pole	240 m/pixel; 6,420 × 6,420 per pole	Light and shadow
South-pole shadow map	Barker et al. (2023)	20 m/pixel; south of 80°S	Fine shadow detail

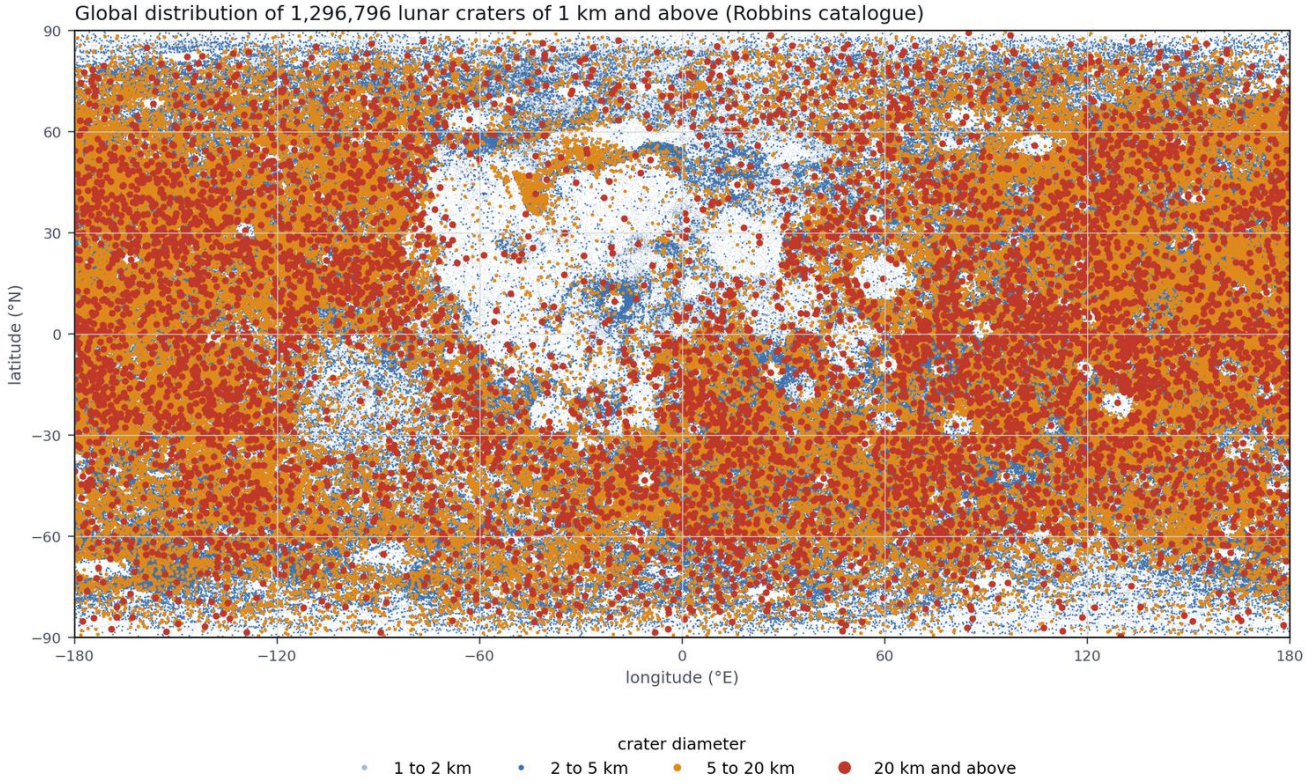


Figure 1. Every crater of 1 km and above in the Robbins catalogue (1.3 million). Dark near-side maria (centre) are sparsely cratered; the highlands are crowded.

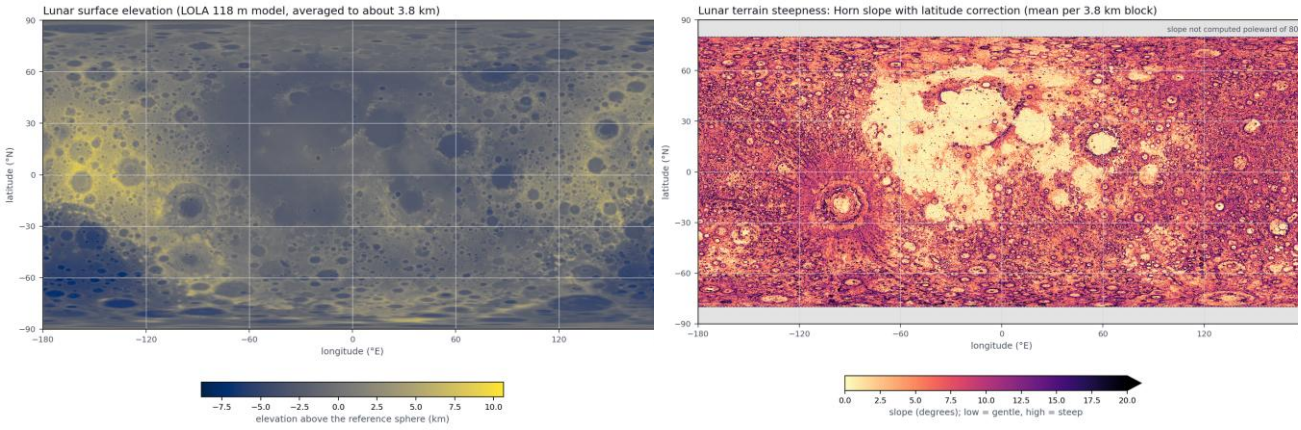


Figure 2. Left: global elevation (118 m model). Right: terrain steepness (Horn slope, latitude-corrected). The smooth maria are gentle; the highlands are steep.

3. DATA PREPARATION (PREPROCESSING)

Raw data from different sources never line up on their own. Most of the effort in this project went into cleaning the layers and joining them correctly, and every step was checked before the next one was built on it.

3.1 Cleaning and standardising the crater catalogues

All three catalogues were converted into the same format: longitude, latitude and diameter (km). Robbins uses longitude from 0° to 360°, so it was converted to the −180° to 180° range to match the elevation map. Otherwise, every crater east of 180° would land on the wrong side of the Moon. In LU5M812TGT, 370 out-of-range longitudes were wrapped by ±360°. Two methods of measuring crater diameter (width and height) were also compared. Since the height-based measurement matched the Robbins catalogue more closely, it was selected for further analysis. Its confidence score was deliberately not used as a filter because it depends on size, and filtering would mostly remove small craters and bias every size comparison.

3.2 Verifying the catalogues

Before using the newer catalogues, it was checked how well they matched the Robbins catalogue. A crater was considered a match if its centre was close enough and its diameter was between half and twice the Robbins crater's diameter.

Checking the crater size was important because without it, a large crater could be mistaken for a smaller crater within it. With both conditions applied, 74%-96% of Robbins craters were found in the other catalogues. The lowest match rate was for LUC-GRAS200 craters between 1 and 2 km. For matched craters, the diameters generally agreed within 2–3%.

3.3 Elevation and Calculating Slope

Each crater was assigned the elevation of the elevation-map pixel at its centre. All 1,296,796 craters were successfully matched to the elevation map. Slope was computed using Horn's method, which estimates the slope at each pixel using its eight neighbours. One important correction was made for latitude. On this map, a pixel is 118 m wide on paper, but on the ground it is only 118 m ×

cos(latitude) wide. Ignoring this difference makes high-latitude terrain look falsely flat. A test with artificial planes tilted at exactly 10° showed the exact problem below:

Latitude of test	Slope - Uncorrected	Slope - Corrected
30°	8.68°	10.0°
60°	5.04°	10.0°
75°	2.61°	10.0°

Slope is reported only up to 80° latitude because the flat map is too stretched near the lunar poles.

3.4 Tiling the Moon

To compare the pixels fairly, the Moon was divided into 2,592 5° × 5 ° tiles. The actual area of each tile was calculated using the Moon's spherical shape, since the tiles near the poles cover less ground area than those near the equator. The density is always measured as craters per 1,000 km².

3.5 Polar layers and the equal-area grid

The polar illumination maps use a polar stereographic projection (positions are metres from the pole), so each of the 191,401 craters poleward of 65° had to be converted into that system before reading the map. The conversion was then verified by checking every map corner to the metre and then confirming that no crater landed outside the data.

Tiles of equal degrees get narrow towards the poles: the 288 tiles above 80° are 11% of all tiles but cover only 1.5% of the Moon, and a polar tile can touch up to 143 neighbours instead of the usual 16. A hotspot test on this grid flagged every polar tile as hot, which was an artefact of the grid rather than a fact about the Moon. The fix was an equal-area polar grid: 20-km-square cells (4,669 per pole) whose true areas are corrected for map stretching. This completely changed the answer for the South Pole.

4. METHODS AND TECHNIQUES

TECHNIQUE	IDEA
Density mapping	Count the craters in each tile or cell, then divide by the true ground area.
Getis-Ord Gi*	This method identifies areas where a cell and its nearby cells (within 61 km) have unusually high or low crater density compared to the Moon's overall average. High scores indicate hotspots (high crater density), and low scores indicate cold spots (low crater density).
Permutation test	We randomly shuffled crater density values among comparable cells 999 times to check whether the detected hotspots could have occurred by chance. This step was important because neighbouring cells are interconnected. The usual statistical threshold ($Z > 2.58$) incorrectly identified 8–23% of cells as significant hotspots, even when the data had no real pattern.
DBSCAN clustering	Group points that have enough close neighbours, with no preset number of clusters.

Association rules (FP-Growth)	Find labels that occur together more than chance ("if in shadow, then latitude 80–90°"), measured by lift (1 = no link).
K-means	Sort craters into groups by size, slope, elevation and latitude; the silhouette score tells how real the groups are.
Control analyses	Split by near/far side, north/south, elevation thirds and longitude sectors to make sure a pattern is not caused by something else.
Catalogue comparison	Repeat every analysis on all three catalogues and measure the overlap of their hotspots (Jaccard index).
Benchmark	Time the same jobs on one machine and on Spark with 1 to 16 cores, with inputs from 1.3 to 42.8 million rows.

5. APACHE SPARK AND APACHE SEDONA

5.1 Processing

Apache Spark divides a large dataset into partitions and processes them in parallel across many cores or machines. Apache Sedona adds spatial capabilities to Spark, including point and polygon types, spatial joins (which tile contains this crater?), raster reading, and spatial statistics. Together, they make it possible to work with very large datasets such as billions of elevation pixels or catalogues with tens of millions of craters.

The project was carried out on a shared university DGX server. The processing setup used:

- Apache Spark: Local mode with a maximum of 16 CPU cores.
- Driver memory: 32 GB.
- PySpark: Version 4.0.
- Apache Sedona: Version 1.9.1.

To check the accuracy of the results, every analysis was first performed using conventional single-computer tools. The same analysis was then repeated using Spark and Sedona, and the results were compared to verify the results.

5.2 Spark and Sedona steps

STEP	SEDONA / SPARK FEATURE	RESULT VS. REFERENCE
Crater points and tile counts	ST_Point builds geometry; Spark groups by tile	Identical
Crater and tile spatial join	ST_Intersects join and tiles as polygons	Identical after fix
Elevation for each crater	Reading raster; RS_Values reads all craters of a tile at once.	Identical after two fixes
Tile elevation statistics	RS_SummaryStatsAll across 36 band files	Differences ~1e-8 (rounding)
Slope	Own latitude-corrected function run on 36 latitude bands in parallel	Matched to 0.0000003°

Clustering	ST_DBSCAN in polar metres	48 settings were run
Hotspots	The distance-band weights and g_local (Gi*)	Z-scores matched 1e-13
K-means, FP-Growth	Spark MLlib	Ran (21 s and 39 s)

5.3 Lessons learned

- Splitting large files improved processing. The 8.5 GB elevation map was stored as a single file, which took more than 16 minutes to process in a single task. Dividing it into 36 smaller raster files allowed Spark to use all 16 CPU cores.
- At the tile boundaries, crater handling was performed. The ST_Contains function missed 217 craters located exactly on tile boundaries, and replacing it with ST_Intersects and ensuring each crater was assigned to only one tile solved the problem.
- Correctly reading the elevation values. Sedona's RS_Values function starts counting pixels from 0 instead of 1. It also automatically applies the 0.5 scale factor stored in the file. Also, applying the scale factor again accidentally halved all elevation values. This issue was identified by comparing the results with the reference data.
- DBSCAN also reported the distance issue. Sedona's spheroid option uses Earth's ellipsoid instead of the Moon's shape, making calculated distances about 3.7 times larger than expected. To avoid this, clustering was performed using planar distances in a polar stereographic coordinate system, which was measured in metres.
- Lastly, handling the Moon's custom projection. The elevation map uses a custom lunar projection that Sedona could not recognise (SRID 0). Instead of allowing automatic coordinate transformation, pixel positions were used directly to avoid incorrect location calculations.

5.4 Spark's Conclusion

Median run times (seconds) for the two benchmark jobs, tile counts and polar cells:

Craters in input	Single machine	Spark, 16 cores	Spark, 1 core
1.3 million	0.07 / 0.06	0.26 / 0.33	0.31 / 0.39
10.7 million	0.40 / 0.30	0.27 / 0.33	0.86 / 0.86
42.8 million	1.47 / 1.10	0.57 / 0.57	3.87 / 3.09

Spark is not necessarily faster for smaller datasets. Spark's 4- to 5-second startup outweighs its benefits in real catalogues. It starts to win at about 10 million rows (1.5× faster) and is 2.6× faster at 42.8 million, with 5.4 to 6.8× speed-up from 1 to 16 cores. For LunarGrid, the main reason for using Spark and Sedona is their ability to handle much larger datasets, such as higher-resolution terrain maps and larger crater catalogues. The same code can also be extended to run on multiple machines in a cluster.

6. ANALYSIS AND RESULTS

6.1 Catalogues agree on where, not on how many

When counted per 5° tile, the catalogues correlate strongly (0.94 to 0.97), indicating they place craters in the same regions.

CRATERS WITH DIAMETER	ROBBINS	LU5M812TGT	LUC-GRAS200
1 km and above	1,296,796	1,998,113	1,813,169
2 km and above	443,016	696,037	474,339
20 km and above	6,973	7,573	7,171

Above 2 km, LUC-GRAS200 and Robbins nearly agree, while LU5M812TGT lists about 60% more. Without ground truth, we cannot say whether the extra craters are real or false detections.

6.2 Crater density rises towards the poles, especially in the north

On the 5° tiles, median density rises steadily from equator to pole in every catalogue (Robbins: 29 craters per 1,000 km² at 0° to 30°, 65 at 80° to 90°).

For example, in the Robbins catalogue, crater density increased from 29 craters per 1,000 km² near the equator to 65 near the north pole. However, this pattern is partly influenced by the lunar maria. These are dark, relatively young plains found mainly near the equator on the Moon's near side, where fewer craters are visible.

When we used an equal-area polar grid to make a fairer comparison, a clearer pattern emerged:

- North pole: Crater density was 2.0×, 1.95× and 1.67× higher than in the 65°–70° latitude ring across the three catalogues.
- South pole: The corresponding values were only 1.15×, 1.03× and 0.92×.

The northern pattern was consistent across all 24 longitude-sector comparisons. However, the southern region showed a more irregular distribution, without a clear increase towards the pole.

The main finding was that crater density increases much more consistently towards the north pole than towards the south pole, and this pattern is supported by all three crater catalogues.

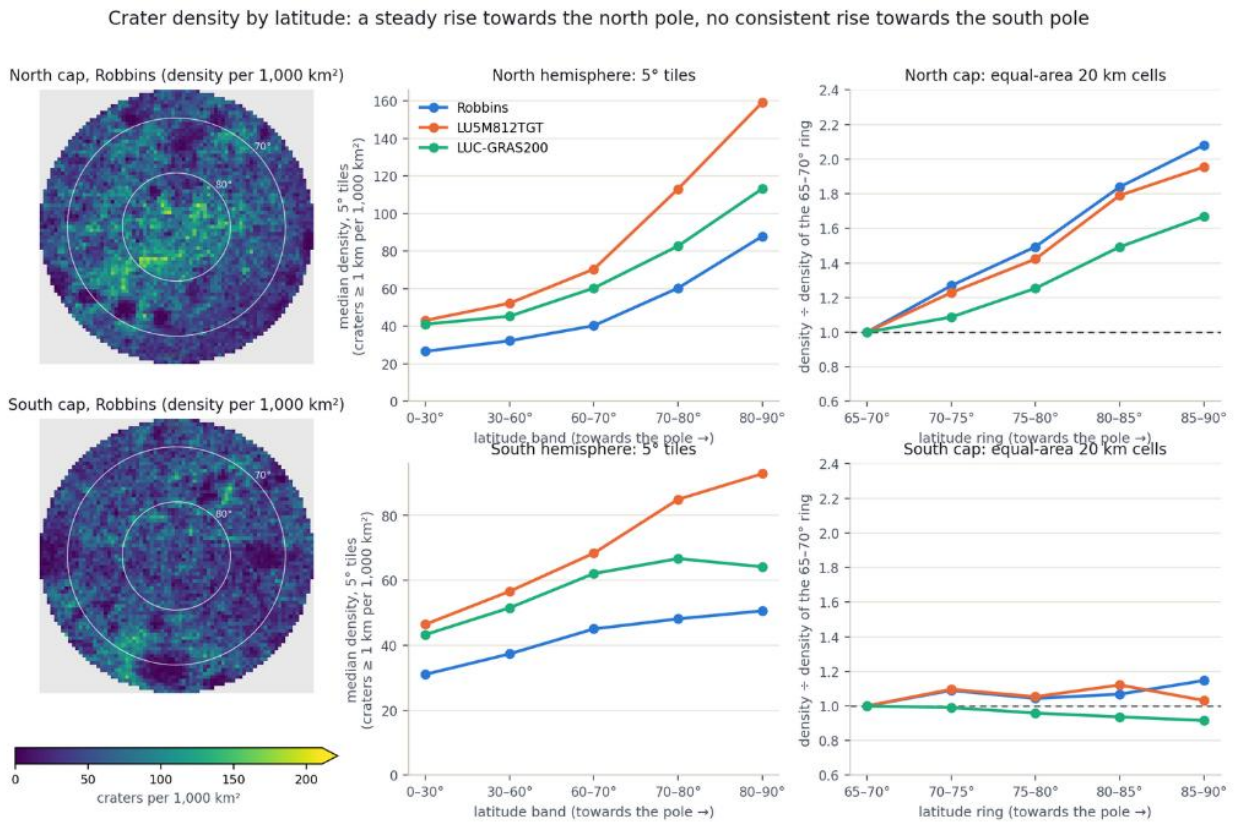


Figure 3. Crater densities by latitude. Left: Robbins density on each polar cap. Centre: 5° tiles. Right: equal-area cells which is relative to the 65°–70° ring. North rises whereas south stays flat.

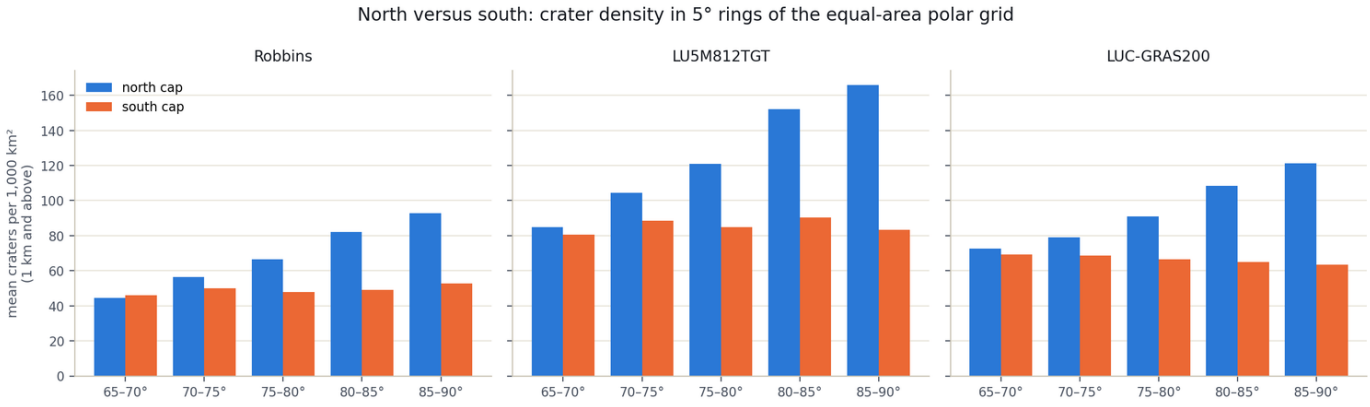


Figure 4. North vs south crater density in 5° rings of the equal-area polar grid, which is noted for all three catalogues.

6.3 Polar hotspots

Getis-Ord Gi* was analysed to identify where dense areas group together. All three catalogues showed the same picture. The most consistent hotspot is in the south, at about 68°S to 73°S and 151°W to 157°W, with density two to five times the cap average; in the north, the strongest cells are near 82°N to 83°N and 34°W to 42°W. The same major hotspot patterns appear in all three crater catalogues, making these locations more reliable for further study.

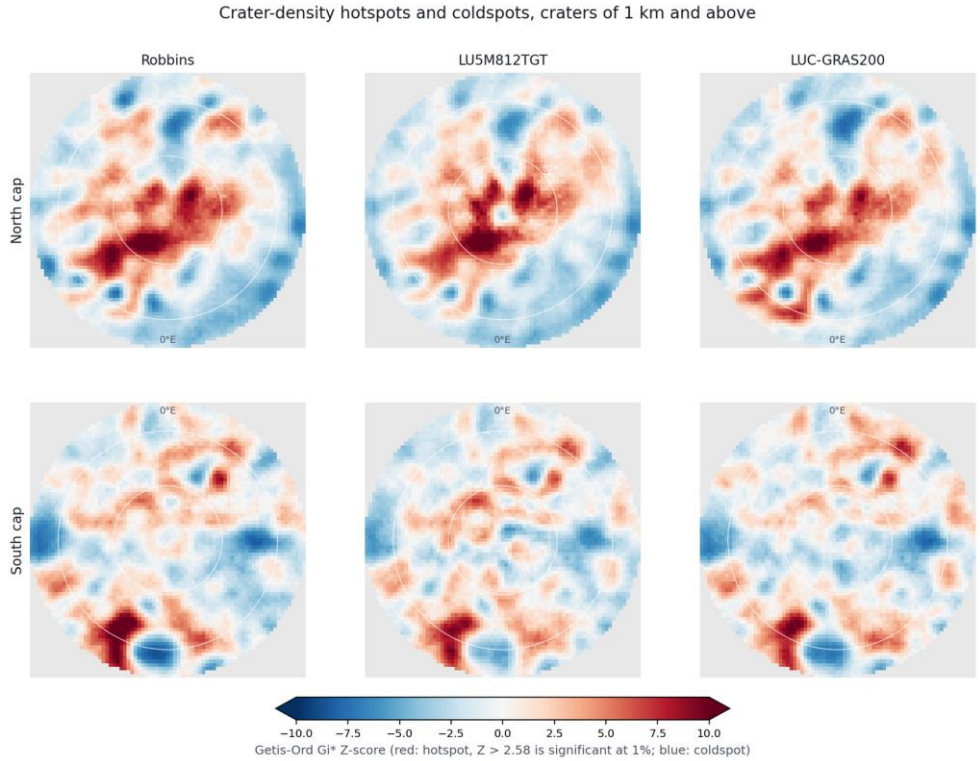


Figure 5. Hotspots (red) and cold spots (blue) of craters ≥ 1 km on each polar cap, for the three catalogues where Z > 2.58

The top 5% of cells overlap far more than chance (Jaccard 0.44 to 0.90, against 0.03 for random sets); agreement is better for craters ≥ 2 km because catalogues disagree most about 1 to 2 km craters. After applying a stricter statistical test, we identified a smaller group of hotspots confirmed by all three catalogues: 36 cells in the north and 158 in the south. These patterns remained almost unchanged even after accounting for illumination differences, suggesting they are not caused by illumination differences.

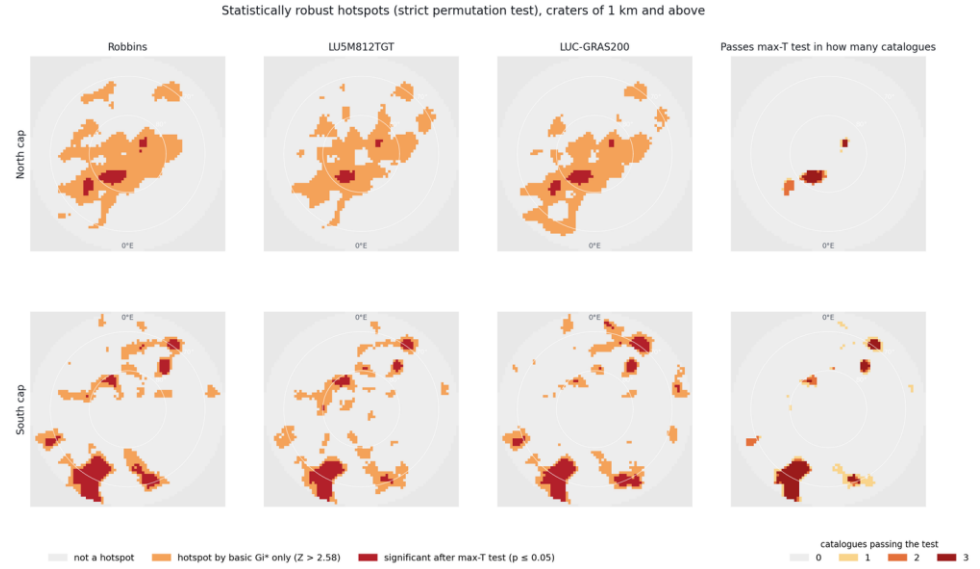


Figure 6. Hotspots surviving the strict permutation test (dark red). Right column: in how many of the three catalogues each cell passes.

The main south hotspot lies in low terrain (median elevation −4,165 m, compared with −1,145 m for other cells) within the South Pole–Aitken basin, the Moon's largest and oldest basin, which fits a heavily cratered, ancient surface, though this geological reading was not tested here.

6.4 The Light and The Shadow

Craters are found in permanent shadow 2.7 to 4.7 times more often than expected by chance. For example, on the 20 m south-pole map, 19.1% of craters are in permanent shadow, compared with only 7.2% of the total area. This is expected because crater floors tend to remain dark, and it also shows that our crater and shadow maps are properly aligned.

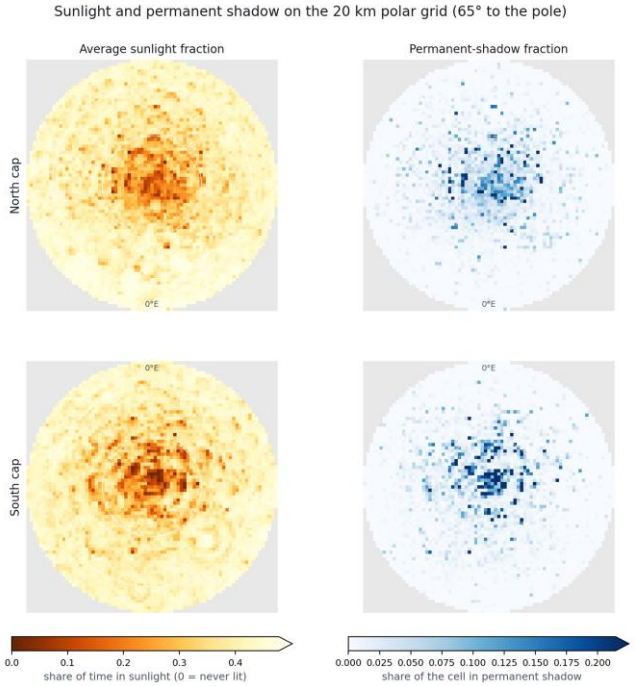


Figure 7. Average sunlight (left) and permanent shadow (right) on the 20 km polar grids. Darker orange means less light; darker blue means more shadow.

Looking at larger 20 km cells gives a more detailed picture. Areas with more than 5% permanent shadow have fewer craters overall than the sunlit areas at the same latitude. However, the pattern changes depending upon the crater size.

Shadowed / lit density	Robbins	LU5M812TGT	LUC-GRAS200
1 - 2 km craters (north)	0.74	0.65	0.72
2 - 5 km craters (north)	0.85	0.76	0.86
10 km and above (north)	1.65	1.50	1.67
10 km and above (south)	2.10	1.72	1.95

Small craters are 25–35% less common in permanently shadowed areas, while large craters are 1.5 to 2 times more common. There are two possible reasons: small craters may genuinely be less common in these areas, or they may be harder to identify in dark images. Also, large craters create their own

shadows, which may partly explain why they are more common in shadowed regions.

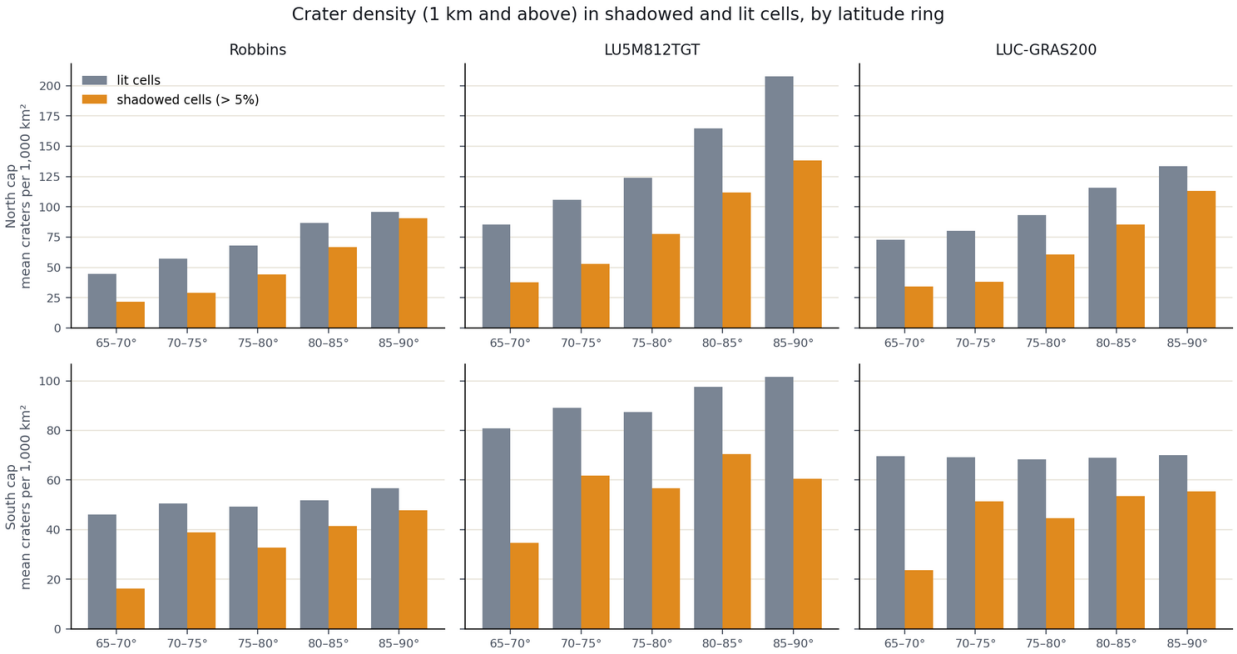


Figure 8. Crater density which is ≥ 1 km in shadowed (orange) and lit (grey) cells by latitude ring. Shadowed cells are less dense, and the gap narrows towards the pole.

6.5 Steepness, clustering and craters' own properties

Slope: Within each latitude ring, the steepest third of cells has 14% to 20% fewer craters than average, mostly because small craters are missing, while large craters make up a bigger share (for Robbins in the north: 21%, 33% and 39% of craters are ≥ 2 km in gentle, middle and steep cells). Since crater walls are steep, large craters partly create the steep terrain, so this is again partly structural.

DBSCAN: Clustering craters near the poles did not give consistent results. With a small search distance (i.e., 2 km), 42–96% of craters were treated as isolated points. With a larger distance of about 10 km, almost all craters merged into one giant cluster. This suggests that craters form continuous patterns rather than separate groups, so hotspot analysis was more useful.

K-means and association rules: K-means clustering identified weak groupings, with a silhouette score of 0.38 for three groups. One group contained larger and steeper craters, with a median diameter of 3.7 km and a slope of 13.8°. This suggests that crater characteristics change gradually rather than forming distinct types.

Association rules mainly confirmed known geographical patterns. For example, about 74% of craters in permanent shadow were located beyond 80° latitude. The connection between shadow and low sunlight is expected, so it is not considered a new finding.

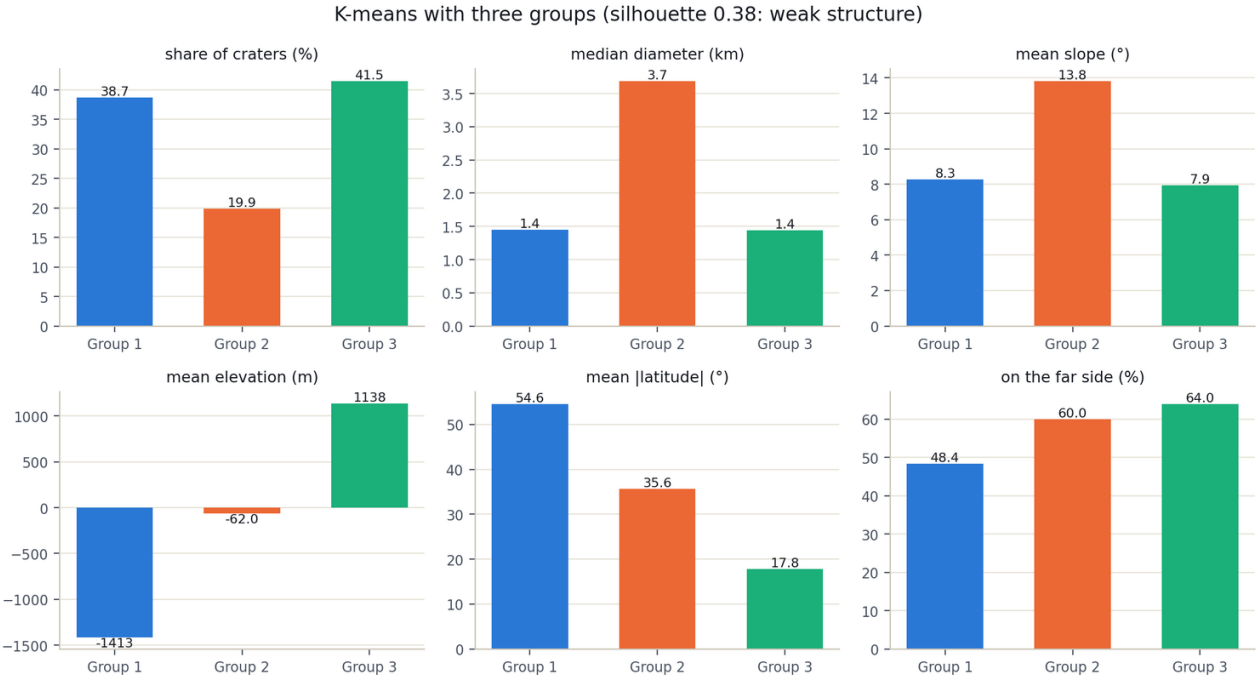


Figure 9. K-means with three groups, i.e. share of craters, median diameter, slope, elevation, latitude and far-side share. A silhouette score of 0.38 shows a weak structure

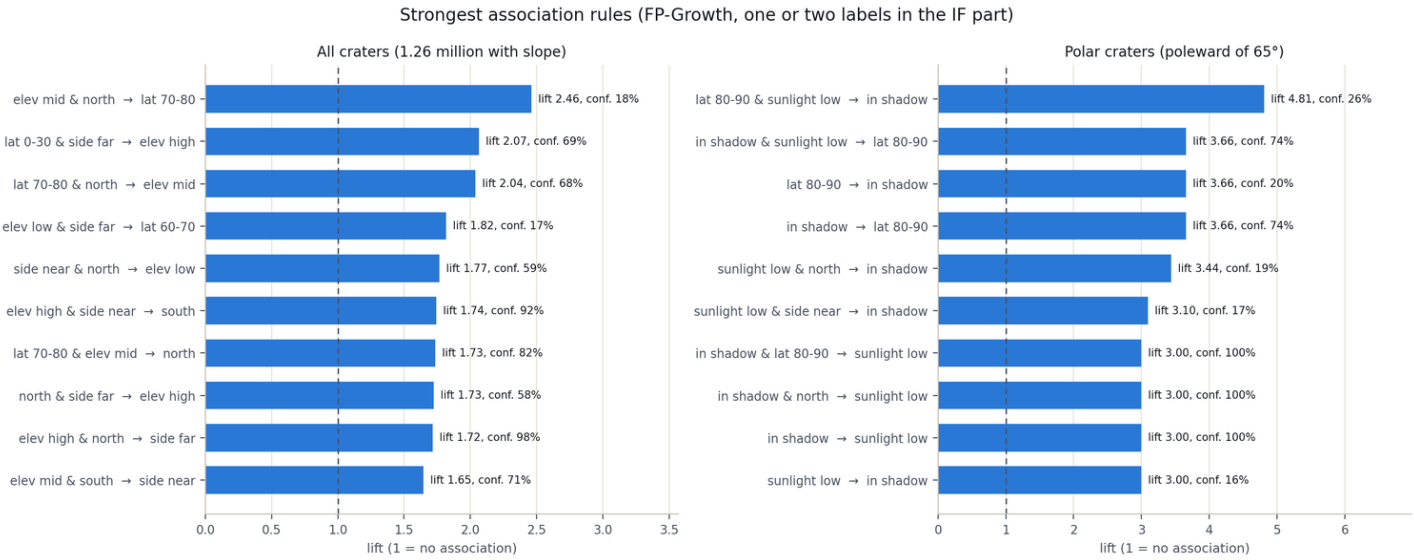


Figure 10. Here, lift; 1 = no association) for all craters (left) and polar craters (right).

7. WHY SPATIAL DATA MATTERS

Almost every conclusion in this project depends on where something is, not just what it is. Below are a few examples that show spatial data is at the heart of the work.

- The same slope reads 10° or 2.6° depending on whether the map's distortion is corrected. The same pole appears to be a hotspot or not depending on whether the grid is equal-area.
- Craters, elevation and shadow have nothing in common except position. Spatial indexing and joins attach terrain and light to millions of craters.
- Ordinary statistics assume independent observations; places near each other resemble each other.

- Spatial statistics (Gi*, permutation tests) are designed for exactly this, and without them a map full of "significant" hotspots would have been reported.
- 5° tiles or 20 km equal-area cells changed the south-pole conclusion, a textbook case of the modifiable areal unit problem.
- Billions of pixels and tens of millions of points need spatial partitioning, which is what Sedona provides on top of Spark.

8. CONCLUSIONS

- Developed a distributed spatial workflow that brings together three crater catalogues, elevation data and polar illumination maps in one consistent framework.
- All three catalogues show similar crater patterns, but their crater counts differ by up to 60% for craters larger than 2 km. Therefore, the results need to be compared across catalogues.
- Crater density increases towards the north pole, but no consistent increase is seen towards the south pole after accounting for latitude, terrain and map distortion.
- Both polar regions have repeatable crater hotspots. This shows that these are not simply caused by sunlight or shadow. The strongest confirmed hotspot region is near 70°S, 155°W.
- Craters are often found in permanently shadowed regions. However, shadowed areas have fewer small craters and more large craters than sunlit areas.
- DBSCAN and K-means did not identify clear crater groups because crater patterns change gradually. Hotspot analysis was more suitable.
- Spark and Sedona become useful for datasets with around 10 million records or more. For smaller datasets, a single machine can be faster, while distributed processing offers better scalability.

9. LIMITATIONS

- The three crater catalogues share some of the same image and elevation sources, and there is no independent ground truth to confirm which catalogue is most complete.
- Crater counts are based on detected locations, but possible detection errors were not included in the hotspot analysis.
- The statistical test identifies clustering but cannot explain its cause. Nearby cells are also related, so the number of truly independent hotspots is smaller than the number detected.
- The terrain and illumination data have limited detail. The 118 m elevation model may not capture smaller craters accurately; slopes are not calculated beyond 80°; shadows are checked only at crater centres; and illumination represents a long-term average.
- Surface types such as mare, highlands and ejecta were not included. The reason for the north–south difference in crater patterns also remains unclear.
- Secondary craters were not reliably identified for the 1–2 km size range near the poles, leaving this question unresolved.
- Performance tests were conducted on one shared machine with cached data. Results may differ on multi-node systems or when reading data from disk. The effects of using different grid sizes were also not tested.

10. FUTURE WORK

- Identifying the areas where the three crater catalogues disagree and checking those locations using satellite images.
- Comparing the craters within similar terrain types to make the statistical tests more reliable.
- Study crater size distributions to understand whether differences are due to surface age or detection limitations.
- Repeat the analysis using different grid sizes to check whether the results remain consistent.
- Use higher-resolution 20 m data for the south polar region.
- Include surface-type maps to distinguish mare, highlands and ejecta regions.
- Test the workflow on a multi-node cluster using GeoParquet or PostGIS for data storage.
- Estimate detection errors and include uncertainty ranges in the final crater density and hotspot maps.

11. KEY REFERENCES

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12. DATA SOURCES AND SOFTWARE

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